Inequalities WS 7

1. a. i. Prove by induction that $(1+c)^n > 1+cn$, for all integers $n \geq 2$, where c is a nonzero constant greater than -1.

   ii. Hence show that $\left(1-\frac{1}{2n}\right)^n > \frac{1}{2}$, for all integers $n \geq 2$.

   b. i. Solve the inequation $\mathrm{x}^2 > 2\mathrm{x}+1$

   ii. Hence prove by induction that $2^{\mathrm{n}} > \mathrm{n}^2$, for all integers $\mathrm{n} \geq 5$.

   c. Suppose that $a > 0, b > 0,$ and n is a positive integer.

   i. Divide the expression $(\mathrm{a}^{\mathrm{n}+1} - \mathrm{a}^{\mathrm{n}}\mathrm{b} + \mathrm{b}^{\mathrm{n}+1} - \mathrm{b}^{\mathrm{n}}\mathrm{a})$ by $(\mathrm{a}-\mathrm{b})$, and hence show that $\mathrm{a}^{\mathrm{n}+1} + \mathrm{b}^{\mathrm{n}+1} \geq \mathrm{a}^{\mathrm{n}}\mathrm{b} + \mathrm{b}^{\mathrm{n}}\mathrm{a}$.

   ii. Hence prove by induction that $\left(\frac{\mathrm{a}+\mathrm{b}}{2}\right)^{\mathrm{n}} \leq \frac{\mathrm{a}^{\mathrm{n}}+\mathrm{b}^{\mathrm{n}}}{2}$.

2. If $a > 0, b > 0$ and $a+b=t$, show that

   a. $\frac{1}{a}+\frac{1}{b} \geq \frac{4}{t}$

   b. $\frac{1}{a^2}+\frac{1}{b^2} \geq \frac{8}{t^2}$

3. Suppose $x > 0, y > 0, z > 0$

   a. Prove that $x^2+y^2 \geq 2xy$

   b. Hence, or otherwise, prove that $\frac{x}{y}+\frac{y}{x} \geq 2$

   c. Prove that $x^3+y^3 \geq xyz\left(\frac{x}{z}+\frac{y}{z}\right)$

   d. Hence show that $x^3+y^3+z^3 \geq 3xyz$

   e. Deduce that $(a+b+c)(a+b+d)(a+c+d)(b+c+d) \geq 81abcd$, where $a, b, c, d > 0$

4. If $a > 0$, show that $a^2+\frac{1}{a^2} \geq a+\frac{1}{a} \geq 2$

5. a. Sketch the function $f(x)=\frac{1}{\sqrt{x}}$

   b. Prove $\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+\cdots+\frac{1}{\sqrt{n}} \leq 2(\sqrt{n}-1)$

6. Let $\mathrm{f(x)} = \mathrm{x}^{\mathrm{n}}\mathrm{e}^{-\mathrm{x}}$, where $\mathrm{n} > 1$

   a. Showthat $\mathrm{f}'(\mathrm{x}) = \mathrm{x}^{\mathrm{n}-1}\mathrm{e}^{-\mathrm{x}}(\mathrm{n}-\mathrm{x})$

   b. Show that $(\mathrm{n}, \mathrm{n}^{\mathrm{n}}\mathrm{e}^{-\mathrm{n}})$ is a maximum turning point of the graph of $\mathrm{f(x)}$, and hence sketch the graph for $\mathrm{x} \geq 0$. (Don't attempt to find points of inflexion)

   c. Explain why $\mathrm{x}^{\mathrm{n}}\mathrm{e}^{-\mathrm{x}} < \mathrm{n}^{\mathrm{n}}\mathrm{e}^{-\mathrm{n}}$ for $\mathrm{x} > \mathrm{n}$. Begin by considering the graph of $\mathrm{f(x)}$ for $\mathrm{x} > \mathrm{n}$

   d. Deduce from part (c) that $\left(1+\frac{1}{\mathrm{n}}\right)^{\mathrm{n}} < \mathrm{e}$

7. Consider the sequence defined by $V_k = \frac{1}{2k+1}+\frac{1}{2k+2}+\cdots+\frac{1}{3k}$, where k is a positive integer.

   a. Show that $V_k < \frac{1}{2}$

b. Given that $p < x < p + 1$, where x is a real number and p is a positive integer. Show that $\frac{1}{p+1} < \int_p^{p+1} \frac{dx}{x} < \frac{1}{p}$

c. Hence show that $\int_{2k+1}^{3k+1} \frac{dx}{x} < V_k < \int_{2k}^{3k} \frac{dx}{x}$

d. Hence find the limit of V_k as $k \to \infty$

8. If $a > 0, b > 0, c > 0$ and $a + b + c = 1$ show that

a. $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$ and $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \geq 27$

b. $ab + bc + ca \geq 9abc$ and $(1 - a)(1 - b)(1 - c) \geq 8abc$

9. Show that $a^2 + b^2 + c^2 \geq ab + bc + ca$. Hence show that $a^4 + b^4 + c^4 \geq a^2b^2 + b^2c^2 + c^2a^2 \geq abc(a + b + c)$.

10. The function $f(x)$ is defined by $f(x) = x - \log_e(1 + x^2)$.

a. Show that $f'(x)$ is never negative

b. Explain why graph of $y = f(x)$ lies completely above the x-axis for $x > 0$

c. Hence prove that $e^x > 1 + x^2$, for all positive values of x.

11. Show that $\int_1^n \ln x \, dx = n \ln n - n + 1$.

a. Use the trapezoidal rule on the intervals with endpoints $1,2,3, \dots n$ to show that $\int_1^n \ln x \, dx \doteqdot \frac{1}{2} \ln n + \ln(n - 1)!$

b. Hence show that $n! < n^{n+\frac{1}{2}} e^{1-n}$. NOT: This is a preparatory lemma in the proof of Stirling's formula $n! \doteqdot \sqrt{2\pi} n^{n+\frac{1}{2}} e^{-n}$, which gives an approximation for $n!$ Whose percentage error converges to 0 for large integers n.

12. The following result applies to any function f which is continuous, has positive gradient and satisfies $f(0) = 0$. $ab \leq \int_0^a f(x)\, dx + \int_0^b f^{-1}(y)\, dy$ (*) where f^{-1} denotes the inverse function of f, and $a \geq 0$ and $b \geq 0$

a. By considering the graph of $y = f(x)$, explain briefly why the inequality (*) holds

b. By taking $f(x) = x^{p-1}$ in (*), where $p > 1$, show that if $\frac{1}{p} + \frac{1}{q} = 1$ then $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$

c. Show that, for $0 \leq a \leq \frac{\pi}{2}$ and $0 \leq b \leq 1$, $ab \leq b \sin^{-1} b + \sqrt{1 - b^2} - \cos a$

d. Deduce that, for $t \geq 1$, $\sin^{-1}\left(\frac{1}{t}\right) \geq t - \sqrt{t^2 - 1}$

Answer

1. b. $x > 1 + \sqrt{2}$ or $x < 1 - \sqrt{2}$;

7. d. $\ln\frac{3}{2}$

12. b.